📝 Assignment 2 (Questions): (Knowing your data, filtering, and sorting)

Supermarket Sales Data Analysis – Filtering, Sorting & Aggregation

Description & Objectives

This assignment evaluates your proficiency in Exploratory Data Analysis using Python Pandas. Working with the `SuperMarket Analysis.csv` dataset (1,000 transactions across three branches), you will demonstrate competency in data ingestion, quality assessment, conditional filtering, multi-criteria sorting, and group-based aggregation. The exercise requires you to extract actionable business insightsincluding revenue patterns by branch/demographic, product performance metrics, and customer behavior indicators—through programmatic data manipulation without modifying source data.

Instructions: Complete the following questions:

Q1. Load the (SuperMarket Analysis.csv) dataset and display the first 5 rows.

In [ ]:

Q2. Display the dataset shape (rows and columns).

In [ ]:

Q3. List all column names.

In [ ]:

Q4. Identify categorical and numerical columns.

In [ ]:

Q5. Check for missing values in each column.

In [ ]:

Q6. Display the data types of each column.

In [ ]:

Q7. Show summary statistics for numerical columns (use pandas method).

In [ ]:

Q8. Filter rows where Sales > 500.

In [ ]:

Q9. Filter sales in City = "Yangon" and Sales > 200.

In [ ]:

Q10. Sort all orders by Sales in descending order.

In [ ]:

Q11. Sort by Date (ascending) and then Time (ascending).

In [ ]:

Q12. Sort by Unit price and Quantity.

In [ ]:

Q13. Calculate the total sales in "Sales" per Branch.

In [ ]:

Q14. Calculate average Sales per City.

In [ ]:

Q15. Find the quantities sold per product line.

In [ ]:

Q16. Calculate average gross income per Gender.

In [ ]:

Q17. Count number of sales per Payment method.

In [ ]:

Q18. Find maximum Sales per Branch.

In [ ]:

Q19. Find minimum Unit price per Product line.

In [ ]:

Q20. Find the sum of gross income per Product line and Branch.

In [ ]:

21. What is the total quantities sold in Product line: "Electronic accessories"?

In [ ]:

22. What is the average Sales for female customers?

In [ ]:

23. What is the most expensive `Unit price` among `Customer type` members only?

In [ ]:

24. How many orders with Rating >= 9 ?

In [ ]:

25. What is the total `Sales` for `Payment` "`Credit card`" in `Branch` C?

In [ ]: